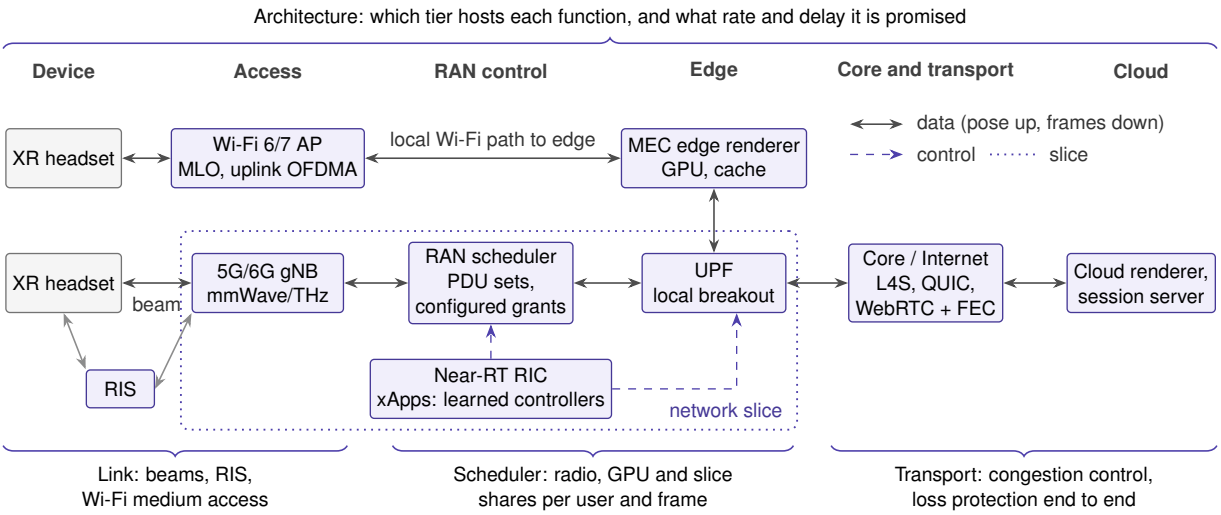


Figure 1: End-to-end 5G/6G and Wi-Fi delivery chain for XR, with the layer at which each subsection makes its decisions. The link acts on the last hop, the scheduler inside the slice, transport across the whole path and architecture across all tiers; most designs optimise one layer and hold the others fixed.

1 Provisioning the Network: Radio, Resources, Transport and Architecture

XR traffic is unlike the traffic that networks were built for. It is *periodic*: the renderer emits a frame every 11–17 ms and the headset reports pose at least as often. It is *bursty*: each frame leaves the encoder as a block of hundreds of kilobytes that must cross the link before the next one [1], and frame sizes fluctuate even under constant-bitrate encoding [2]. It is *bidirectional*: pose, input and, for edge AR, camera frames go up while rendered content comes down. It is *deadline-bound*: a frame that misses its display slot is worth no more than a lost one. Best-effort, throughput-oriented networks assume elastic flows that tolerate queues and value average rate, and XR breaks each of these assumptions. Provisioning must therefore supply rate, delay and reliability per frame, not per flow, and the tail of frame delivery, not its mean, decides whether the user sees a stall. How much traffic there is depends on where computation is split (**??**): a remote renderer sends video, a local renderer sends scene state. This section works bottom-up (Figure 1): the link, the scheduler that shares it, the transport that crosses shared bottlenecks, and the architecture that places functions and states what the network must promise.

1.1 The link: a directional channel on a moving head

The last hop must carry multi-gigabit bursts to a head that turns and is easily blocked, and its tail, not its peak rate, is the limit. Approaches fall into four families.

Directional links (mmWave, THz) buy rate with narrow beams and pay in fragility. THz analysis gives 99.999% reliability at 20 ms with guaranteed line of sight, but with blockage reliability falls to 96% at 20 ms and tail delays reach 30–90 ms while the mean stays below 20 ms [3]. At 60 GHz, 3D head motion defeated beam search designed for 2D [4], and commercial hardware lost up to 4.72% of packets under 45°/s rotation [5]. Blockage and rotation, not physical rate, limit these links.

Diversity and pose-steered beams are the remedy that holds across studies. Dual-beam reception with coordinated access points raised reliability by up to 22.8% on real headset motion [6], and distributed antennas kept 12 Gb/s under full rotation [7]. Steering beams from head orientation cuts search overhead [8]; predictive beams from a neural radio field halved beam-search error on a 28 GHz array [9]. These designs reuse the motion predictors of **??**, so prediction error bounds their gain.

Reconfigurable surfaces (RIS) promise to route around blockers, but the promise is assumed. RIS-assisted XR has never run on hardware, learned RIS control violated a 20 ms bound beyond 15 simulated users [10], and optimisers that take seconds per slot [11], or minutes on practical prototypes [5], must react within a head turn.

Shared medium access binds on Wi-Fi and 5G before physical rate does. With frames aligned to the 802.11ad beacon interval, eight headsets at 400 Mbit/s met a 1 ms bound [12]. Wi-Fi 7 multi-link

Table 1: Families of XR schedulers. Each row names what the scheduler decides, what it takes as given, and where it wins and fails.

Family	Decides	Assumes	Strength	Limit
Optimisation (convex, Lyapunov, ADMM) [18, 21, 22]	Cache, render and radio shares per slot	Known cost, QoE and channel models	Bounds; near-optimal at low run time	Breaks under model mismatch; rendering cost assumed
Heuristics (greedy, approximation) [23]	Per-user quality level and share	Monotone utility	Simple; provable fraction of optimum; runs on devices	No delay guarantee under bursts
Learning (DRL, bandits, xApps) [24–27]	Admission, resource blocks, slice size	Test traffic resembles training traffic	Adapts to unmodelled dynamics	Weak baselines; generalisation and inference cost untested
XR-aware mechanisms (PDU sets, XR DRX, CBG-HARQ) [17, 28–30]	Frame-level delay budget, sleep, retransmission	Standard XR traffic model	Frame semantics in the RAN; deployable	System-level simulation only

operation served 7 rather than 6 AR stations at 80 MHz [13], and doubling access point antennas added only 35% capacity [14]. Uplink OFDMA with three headsets disrupted tracking on real hardware [15], and whether uplink or downlink binds is contested because it depends on the assumed rates [13, 16]. On 5G-Advanced, code-block-group retransmission with outer-loop link adaptation reports up to 67% more satisfied XR users [17], in system-level simulation only.

1.2 The scheduler: deciding per frame who gets what

A scheduler divides spectrum, power, edge GPU time and cache among users at every slot. These resources substitute for each other: caching rendered fields of view and computing on the device can stand in for bandwidth [18, 19], and joint uplink, downlink and edge allocation saves mobile energy in collaborative AR [20]. The tension is between the model a scheduler optimises and the traffic it meets. Table 1 groups the approaches into four families.

Optimisation wins when its models hold: multicasting overlapping 360-degree viewports over mmWave under a Lyapunov delay constraint raised HD delivery by 40–50% and cut 99th-percentile delay by 17–37% against unicast, in simulation driven by 50 real head traces [21]; ADMM placed objects for edge VR games four orders of magnitude faster than a commercial solver [22]. *Heuristics* trade guarantees for deployability: a collaborative-VR allocator reached at least half the optimum and ran on commodity phones [23]. *Learning* schedulers beat greedy, random and proportional-fair baselines [24, 25], but these baselines are weak, and learning and optimisation are rarely compared on the same traces. On O-RAN testbeds the trade-off becomes concrete: latency-estimation control met 10–16.67 ms targets but over-provisioned, packet-size prediction was efficient but lost latency control [31], and a latency-steering xApp freed 15.8% more resources than an equivalent static slice [27].

XR-aware mechanisms change what the radio sees rather than how it optimises. A PDU set groups the packets of one frame so that the RAN can drop or prioritise them together under one delay budget [28]; a delay-tracking PDU-set scheduler served 4–5 times more satisfied users than earlier PDU-set-aware schedulers under scarce resources [29]. Discontinuous reception aligned to the frame period saves headset power [30, 32], and configured grants could pre-allocate the periodic pose uplink. The uplink itself is rarely scheduled for tracking or camera data [24, 33], and no testbed has compared configured with dynamic grants.

Three problems sit above the family choice. *Joint radio–rendering allocation* treats GPU capacity as a free variable [34, 35]; only one testbed adapted base-station bandwidth and edge GPU frequency together [36], and the one measured GPU figure is economic, a 4 GB vGPU slice that halved cloud-XR hourly cost while holding MOS at 4.38 or above [37]. *Multicast* gains depend on viewport overlap [11, 21, 38], so they are unproven for interactive 6DoF content, although one Wi-Fi access point and an edge server served volumetric video to over 30 untethered devices [39]. *The objective* is contested: schedulers optimise a utility of tracking and transmission delay [24], SSIM and VMAF regressions [40],

Table 2: Stated requirements against measured or derived figures. The gaps follow from unstated content representation, codec and control loop.

Quantity	Stated	Measured or derived
Per-user metaverse rate	100–250 Mbps [54, 64]; $\sim 10^3\times$ 5G throughput [56]	Horizon Workrooms 2–3 Mbps down, 0.6 Mbps up, +0.5 Mbps per extra user [62]
Holographic call	1–10 Tbps; 6.63 TBps for a light-field display [65]	5–100 Mbps for a compressed point cloud over live 5G [66]; $<$0.2 Mbps for a driven avatar [67]
Untethered VR link	15.75–26.25 Gbps uncompressed [5]	400 Mbit/s per 4K headset, eight per 802.11ad AP [12]
Interaction latency	1 ms for tactile and telesurgery use [55]	Haptic users tolerated 200 ms with adaptive control [68]; cloud VR RTT up to 90 ms hardly noticed [42]
Discomfort threshold	30 ms [64] vs $<$20 ms [55]	Not derived from a user study in either source

an attention-weighted "Meta-Immersion" score [34] or weighted sums of quality and delay [23]. Gains of 20–37% are reported in each paper's own metric, none validated with users (**??**). Traffic inputs are optimistic too: the 3GPP baseline model underestimated effective VR bitrate by 25% against a measured headset [41].

1.3 Transport: congestion and loss for interactive frames

End to end, frames cross shared bottlenecks where loss-based congestion control fills queues, and a retransmission usually arrives after the frame's deadline. Users find loss the greater threat. In cloud VR, RTT up to 90 ms was hardly noticed while loss up to 8% mattered [42]; with CloudXR, QoE fell at RTT of 75 ms or loss above 6% (N=30) [43]; with Air Link, loss raised cybersickness more than delay did (N=16) [44]. Short stalls of up to four frames at 90 fps went unnoticed, and reprojection helped from eight frames upward [45], which gives transport a margin that the compute pipeline sets (**??**). Three families address the problem.

Low-queue congestion control keeps the bottleneck short. An L4S pipeline for volumetric conferencing converged 45% faster than Google congestion control and lost no frames where WebRTC lost 3% [46], and L4S has been carried over 5G for split rendering [47]. Both used controlled bottlenecks with one competing scheme; radio links with fast capacity swings are untested. *Session protocols* decide how media and control share a connection. Practice is ahead of research: Steam Link carries video and control in one QUIC tunnel [48], Media over QUIC, RTP over QUIC and WebRTC have been compared for holographic conferencing over Wi-Fi and 5G [49], and a WebRTC selective forwarding unit adapted 13–36% faster than TCP for volumetric streams in emulation [50]. On the air, WebRTC batching of frames at fixed intervals wasted Wi-Fi airtime [51], while aligning frame delivery with the display's VSync removed up to 14.3% of frame misses [1]. *Loss protection* repairs damage without waiting for a round trip. Intra refresh reduces latency variability on Wi-Fi [52], but forward error correction for remote rendering has barely been studied [41, 53], although radio losses are bursty and users weigh them most.

1.4 Architecture: where functions sit and what the network must promise

Architectures decide which tier hosts rendering, state and caching, and what the network must deliver. Four families recur. *Edge-centric* designs make edge computing a necessary tier, an agreement shared across surveys [54–56]. *Sliced* designs isolate XR traffic; this family has testbed evidence, since AR, cloud gaming and video calls were measured under RAN-slice configurations on an O-RAN 5G network [57]. *6G and metaverse visions* add THz access, AI-native control and ledgers [56, 58], but rarely map their functions onto the 5G-Advanced XR mechanisms that already exist [59, 60], so what 6G must add remains unstated. *Measured platforms* tell a plainer story: throughput, latency and device load on five social VR platforms grew almost linearly with users because servers forward avatar data directly [61, 62], and locality-based server assignment scaled VR conferencing to hundreds of users [63]. The scaling bottleneck there is server logic, not access capacity (**??**).

Requirements need the same discipline as results. Architecture papers agree on the qualitative

list but not on the numbers, and Table 2 shows why: stated per-user rates exceed measured ones by two to five orders of magnitude because the operating point is left unstated. An uncompressed light field is not a compressed point cloud, and 360-degree video is not avatar state. A requirement means something only with its representation, codec, split point and control loop attached. The figures also propagate: one survey reproduces another's requirement table with a reliability target changed from $1 - 10^{-6}$ to $1 - 10^{-5}$ [54, 69], so the apparent consensus is partly one source counted several times. One architectural claim remains untested: ledgers are placed in the interactive data path for ownership and trust [54, 58], yet no study has measured the delay they add, although one concedes that replication raises delay with users [70].

Open problems

P1. Which objective should an XR scheduler optimise? Allocators optimise incompatible QoE constructs that no user study has validated. A QoE-in-the-loop slicing study on an open 5G testbed, comparing static, latency-target and QoE-model xApps with ratings and sickness scores, would decide which objective matters.

P2. Do joint radio–rendering allocators survive measured costs? Rendering capacity is a free variable in these models. Profiling render and encode time and GPU power on edge hardware, then re-running published allocators under measured costs, would show whether their rankings hold.

P3. Can RIS or THz links meet frame deadlines for real XR traffic? RIS work is simulated and THz reliability rests on Poisson traffic. A pose-driven RIS prototype at 28 or 60 GHz, measured on frame deadline misses under head motion, would test the premise.

P4. How should the uplink be granted? Pose and camera uplink enter motion-to-photon latency directly, yet configured, dynamic and pose-prioritised grants have never been compared on a testbed with tracking error as the outcome.

P5. How should congestion control and loss protection share a frame deadline? Users weigh loss above delay, but L4S and forward error correction are tested only on emulated bottlenecks. Comparing them over cellular and Wi-Fi with cross traffic, reporting 99th-percentile frame latency, would settle the design.

P6. What do shared worlds actually require? Requirement tables are copied, not derived. Traffic measurements of shared-world platforms across user counts and avatar fidelities, compared with standard XR traffic models, would replace them.

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